Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Too Good To Be True
Project / Solution Name	eStock
Name (Leader)	Marco Ryan Suseno
Email & Phone (Leader)	hyprmarctheprime@gmail.com / +60 11-2561 2588
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://e-stock-maximillionc.replit.app
Code Repository URL (public / access granted to judges)	https://github.com/Swiftny/eStock

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

eStock uses a React/TypeScript interface served by Next.js, with server routes running on Node.js. Replit is used for application deployment. Browsers communicate with the backend rather than directly accessing PostgreSQL or receiving API credentials.

A Manager uploads a supplier-document image. The backend calls Gemini 3.8 Flash, validates its structured response and matches descriptions against PostgreSQL catalogue names and confirmed aliases. Manager corrects the draft and sends a frozen receipt without changing stock.

Shared Staff checks physical goods and enters actual quantities. The backend derives shortages or extras and commits actual stock, movements, receipt completion and required Manager notifications in one transaction. Both roles also perform validated Take/Return operations.

PostgreSQL stores products, aliases, drafts, receiving results, movements, notifications, request identities and hashed sessions. Permissions, revisions and locking protect operations. Docker provides separate local development databases. Earlier RunPod forecasting training is outside the current receipt path. Appendix A shows the connections.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 6.0.3	Application types, frontend and backend development.	Apache License 2.0
Next.js 16.4.2	Web application framework, routing and server-side endpoints.	MIT

Name & Version	Purpose in Project	License
React and React DOM 19.3.1	Interactive Manager and Staff interfaces.	MIT
Node.js; project minimum 22.18.0	Server runtime and development tooling. Exact deployed version requires confirmation.	MIT, with bundled component notices
Google Gen AI SDK, @google/genai 2.31.0 declared dependency	Server-side integration with the Gemini API.	Apache License 2.0; Gemini service use is separately governed by Google's terms
node-postgres, pg 8.23.1	PostgreSQL connections, SQL operations and transactions.	MIT
PostgreSQL 17; local image postgres:17-alpine	Local development database and relational persistence. The hosted database version requires confirmation.	PostgreSQL License; container components retain their respective licences
@fontsource-variable/manrope 5.3.0	Website typography.	Manrope font: SIL Open Font License 1.1
Biome 2.5.15	Formatting and static code checks.	MIT OR Apache License 2.0
playwright-core 1.64.1	Automated browser verification.	Apache License 2.0, including applicable notices
TypeScript @types packages	Development type definitions for Node.js, React and PostgreSQL.	MIT
Docker Compose; installed version not recorded	Reproducible local PostgreSQL setup on individual laptops.	Apache License 2.0

Name & Version	Purpose in Project	License
Docker Desktop; installed version not recorded	Local container management.	Docker Subscription Service Agreement and component licences
Python; experiment interpreter version not recorded	Historical forecasting preparation, evaluation and supporting scripts.	Python Software Foundation licence and applicable distribution notices
AutoGluon TimeSeries 1.6.3 — historical	Earlier demand-forecasting training and evaluation.	Apache License 2.0
chronos-forecasting 2.3.2 — historical	Earlier Chronos forecasting experiments.	Apache License 2.0
PyTorch 2.13.0+cu130 — historical	GPU computation during earlier forecasting experiments.	BSD-style licence; applicable dependency and CUDA terms
RunPod Python SDK 1.12.0 — historical	Supporting earlier RunPod experiment operations.	MIT

The current application uses SQL through node-postgres. No ORM, retrieval-augmented generation system or vector database is required by the current receipt workflow.

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Google Gemini API — Google	Reads uploaded supplier-document images and returns structured candidate fields, including product descriptions, quantities and dates. The current configured model identifier is gemini-3.8-flash.	Free

Gemini is called from the backend. API credentials are not sent to the browser.

Catalogue matching, permissions, receiving validation, stock calculations and database updates are handled by application logic. Gemini does not directly write inventory balances.

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Application workspace and deployment of the eStock web application. Public deployment: https://e-stock-maximillionc.replit.app	Sponsor Provided
Hosted PostgreSQL service	Database required by the hosted application. Local development uses PostgreSQL in Docker.	Free

Service & Provider	Purpose in Project	Plan / Region
RunPod — historical GPU experiment	GPU compute for an earlier forecasting prototype. The inspected proof shows the Pod named estock-train using one NVIDIA RTX 4090. This is separate from the current Gemini receipt workflow.	Sponsor Provided Region: EUR-IS-2 1x RTX 4090 Secure Cloud
GitHub.	Public source repository and version history.	Free

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 3.8 Flash — gemini-3.8-flash	Google Gemini API	Pre-trained multimodal model configured for supplier-document extraction. No team fine-tuning or self-training of this model. Successful operation of the final model on the final hosted deployment still requires acceptance testing.	Google Gemini API Terms of Service and applicable account terms: https://ai.google.dev/gemini-api/terms
Gemini 2.5 Flash — gemini-2.5-flash	Google Gemini API	Historical exploratory OCR evaluation. Also remains the source-code default when no model override is supplied. The inspected local configuration overrides it with gemini-3.8-flash.	Google Gemini API Terms of Service and applicable account terms

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Chronos-2 — autogluon/chronos-2, with a historical LoRA fine-tuning experiment	AutoGluon / Chronos ecosystem; model source: https://huggingface.co/autogluon/chronos-2	Earlier demand-forecasting prototype trained/evaluated using RunPod GPU compute. It is not part of the current receipt-extraction or receiving path.	Base model and relevant forecasting tooling: Apache License 2.0; applicable dependency terms also apply

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic inventory, receiving and transaction fixtures	Team-created repository fixtures and isolated automated tests	Synthetic	Team-created project material; applicable project licensing and attribution	N
Synthetic supplier-document test images	Team-created document fixtures used for controlled extraction and browser testing	Synthetic	Team-created project material; no external dataset licence claimed	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Private supplier-document photographs used for exploratory OCR QA	Team-provided local reference collection; 122 retained JPEG files, with a selected subset submitted to Gemini during exploratory testing	Private / team-provided	No public dataset licence established. Permission, retention and competition-data review remain unresolved.	N
sales_history.csv	Team created using data/generate_history.py; see data/DATA_CARD.md	Synthetic	Apache License 2.0 (eStock repository)	N
estock_sales_daily_5sku_2020-01-01_to_2026-01-01.csv	Team-provided file; see data/DATA_CARD.md	Synthetic	No separate dataset license established	N

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
OpenAI Codex — GPT-6.1 Sol	Application implementation, debugging, tests, code inspection, integration and technical documentation.	Major
ChatGPT	Brainstorming, problem refinement, requirements, prompts, interface wording, mentoring preparation and submission drafting.	Major
Claude Opus 5.5	Assistance with selected implementation and development tasks, followed by team integration and review.	Moderate
Replit Agent	Brief configuration and deployment-related assistance.	Minor

Supporting development and reference tools

Context7 / ctx7 and documentation-retrieval tooling for technical references.

Google Drive for organiser materials and submission templates.

UI/UX Pro Max and Anti Slop instruction sets for interface guidance and quality review.

Git, GitHub, Visual Studio Code, Python tooling, browser developer tools and Playwright-based checks.

Attio’s public website as a visual reference for restrained layouts, typography and information hierarchy. No proprietary Attio code or assets are claimed as team-created material.

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-assisted changes were checked using TypeScript, Biome, production builds, automated PostgreSQL/API tests and isolated browser journeys. Retained reports cover permissions, quantity validation, concurrent operations, repeated submissions and transaction rollback. Synthetic provider stubs are distinguished from exploratory live Gemini calls. Product requirements and corrections were directed by the team. Members remain responsible for reviewing their contributions, verifying resource declarations and explaining the implementation. Final sign-off must confirm the submitted build, data permissions, public access and outstanding limitations; automated checks do not establish broad OCR accuracy or complete production readiness.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
Replit	Application platform and deployment facilities	Used for the eStock workspace and public web deployment. The exact plan and whether sponsor credits were redeemed require account confirmation.
RunPod	GPU compute facilities and account credits	Used for the earlier forecasting experiment on the estock-train Pod with one RTX 4090 in EUR-IS-2. The precise source of account credits requires confirmation.
Alpha Red Solutions / Orbyte	Domain materials and mentorship	Domain 2 materials and mentoring feedback from Cheah Kwan Cheong informed problem refinement, especially practical workflow fit, repeated data entry and duplication with existing inventory systems. Final product decisions and implementation remained the team's responsibility.

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit provides application hosting and is retained to meet the hackathon's platform requirement. Another compatible host could technically replace it, but that alternative has not been tested. RunPod enabled earlier GPU forecasting experiments; the current receipt workflow does not require that model or a RunPod endpoint. Alpha Red mentoring informed problem refinement, while the team made final decisions. Gemini supplies current extraction and would require fresh prompt, schema, accuracy and privacy validation if replaced.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

eStock stores catalogue, receipts, quantities, movements, notifications, request identities and role/session metadata in PostgreSQL. Runtime images are sent in full to Google Gemini, including visible personal or payment details ignored during extraction. Runtime uploads are transient; drafts retain structured fields.

Separately, 122 private QA image files are retained locally outside public Git tracking. They contain real personal data; selected images were sent to Gemini. Authorisation and deletion arrangements require review.

API keys stay server-side. Permissions, same-origin checks, hashed session tokens and HttpOnly/SameSite cookies restrict access; production cookies are Secure. Sessions expire after 24 hours; logout deletes the current session, but expired rows are not automatically purged.

Structured records persist until authorised clearing; no automatic retention schedule is configured. Provider/hosting retention, regions, logs and backups remain unconfirmed. Shared Staff does not identify individual employees.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Gemini may misread names, dates, quantities, units or document types and return plausible errors. Manager reviews extraction, catalogue matches and manual unit conversions; Staff confirms physical quantities before stock increases. An image cannot prove delivery.

Transactions, frozen revisions and request identities protect receipt posting. Separate uploads can still create duplicate drafts; lost responses require checking Drafts before retrying. Provider HTTP 503 receives at most one extra attempt. Other provider, network and hosting limits can interrupt processing; there is no fixed application extraction deadline.

Earlier live samples were small and sometimes slow. They do not establish accuracy or performance for the final model/deployment. Physical devices, authenticated hosted workflows and restart persistence remain unverified. Shared Staff limits individual accountability; deliberate History clearing removes detail. Real personal data in private QA references creates unresolved permission, retention and competition-data concerns.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- [v] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [v] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [v] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [v] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [v] Any use of official SDC sponsor resources has been declared in Section 5.
- [v] The prototype described here is accessible to judges through the links provided in Section 1.
- [v] All fields in this template have been completed within the word limits stated.

8. Declaration

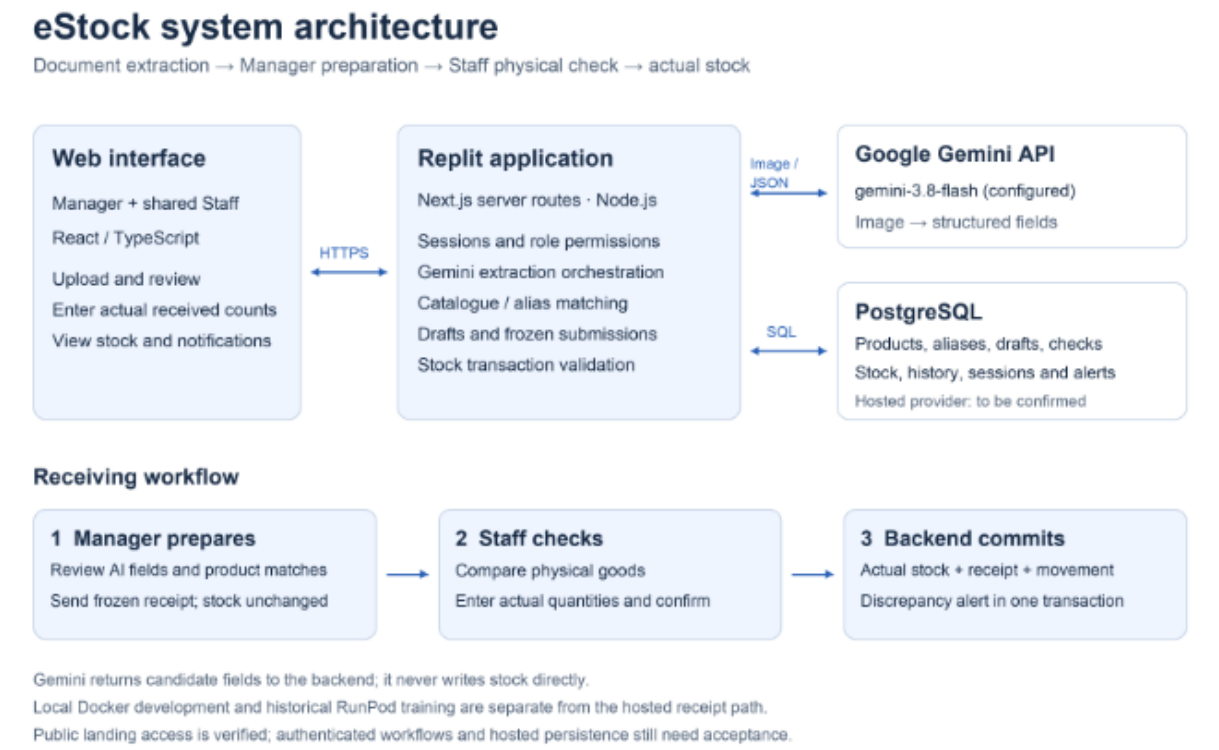
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Name: Marco Ryan Suseno

Signature: 

Date: 25 – 10 – 2026

Appendix A - System Block Diagram



The diagram shows:

Browser interface → Replit-hosted Next.js backend → Gemini extraction API → backend validation and catalogue matching → PostgreSQL.

The receiving sequence is:

1. Manager uploads a document and reviews extracted details.
2. Manager sends the prepared receipt to Staff. Stock remains unchanged.
3. Staff checks the physical delivery and enters actual received quantities.
4. The backend derives matching, short or extra quantities.
5. Actual stock, receiving completion, stock movements and required Manager discrepancy notifications are committed together.

Gemini returns candidate document fields. It does not independently confirm physical deliveries or directly update stock.

Local Docker databases and historical RunPod training are separate from the hosted receipt-processing path.

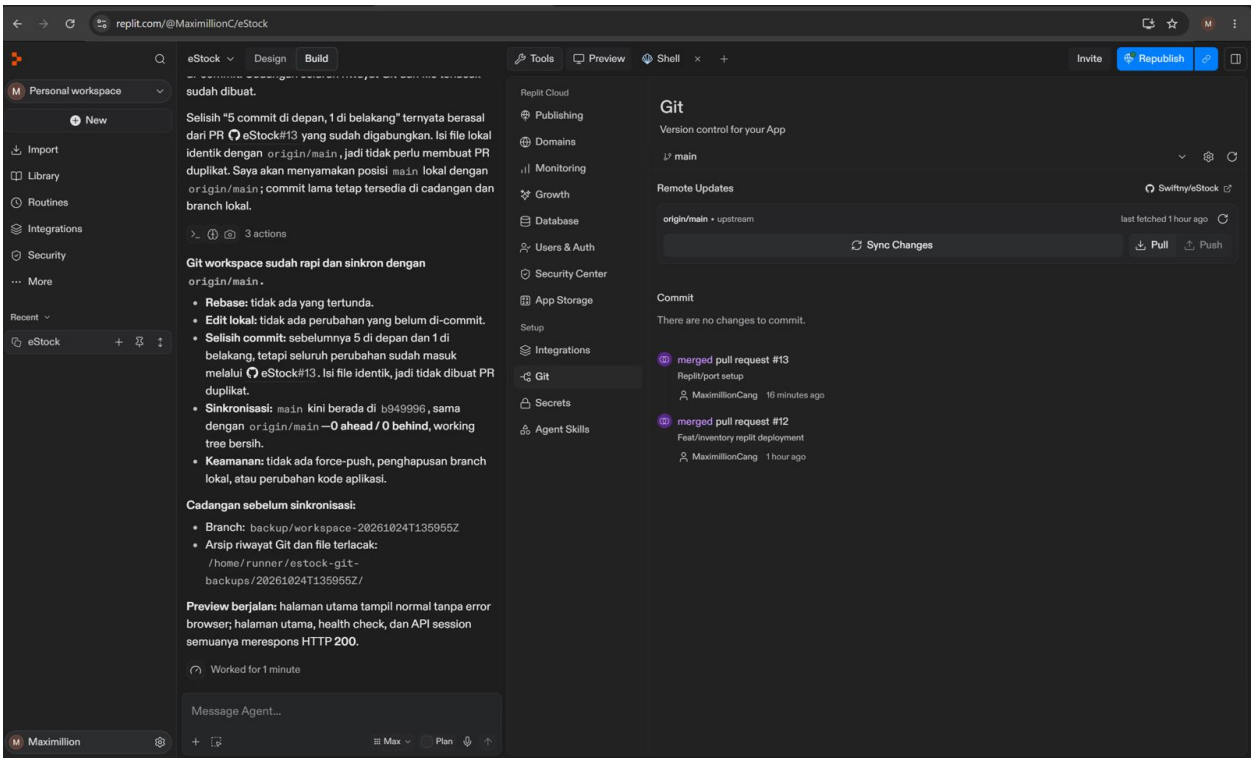
Appendix B — Replit Proof of Use

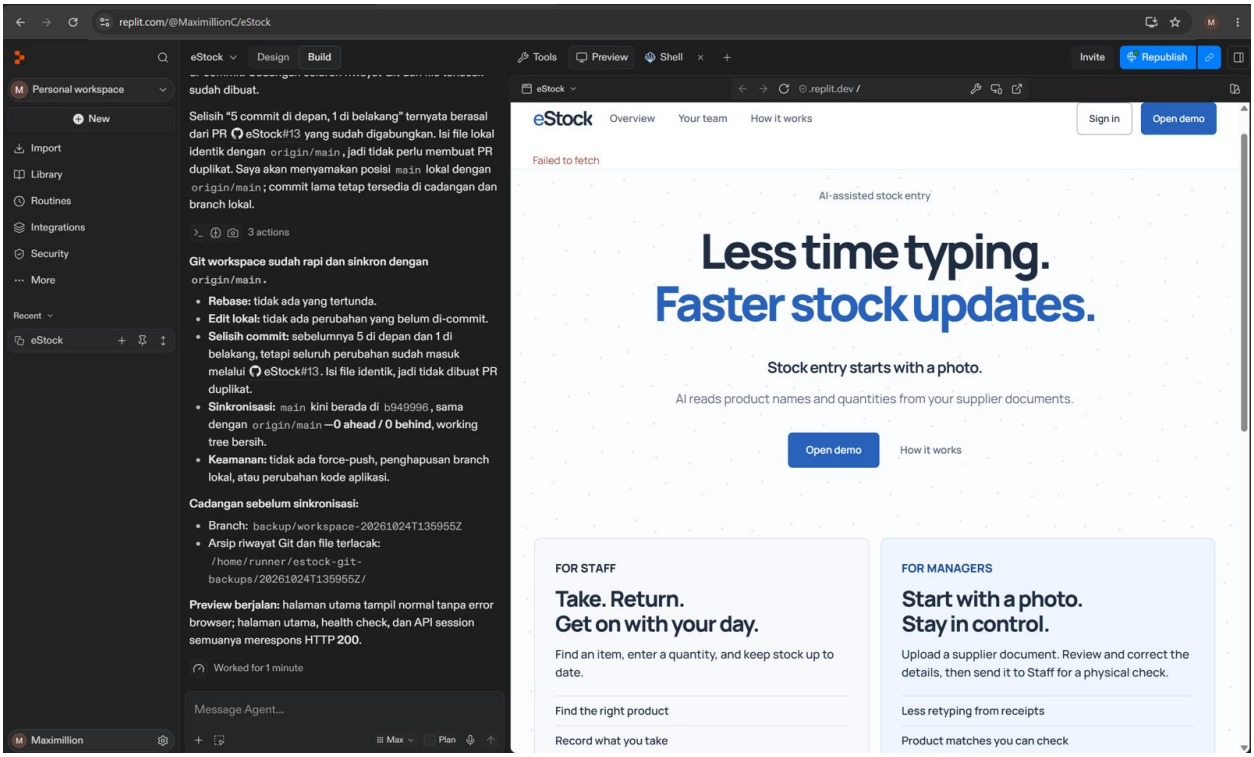
Public application:

https://e-stock-maximillionc.replit.app

Source repository:

https://github.com/Swiftny/eStock





The inspected proof establishes Replit workspace use.

Appendix C — RunPod Proof of Use

Purpose: Historical forecasting experiment preceding the current receipt-based workflow.

Pod name: estock-train

GPU: 1 × NVIDIA RTX 4090

Region: EUR-IS-2

Pod ID: rwqsww7pn19mcu

Base model: autogluon/chronos-2

Recorded base-model revision:

60088152a34e242427b44c3100014473a0157d53

Experiment: LoRA fine-tuning / forecasting evaluation using AutoGluon TimeSeries.

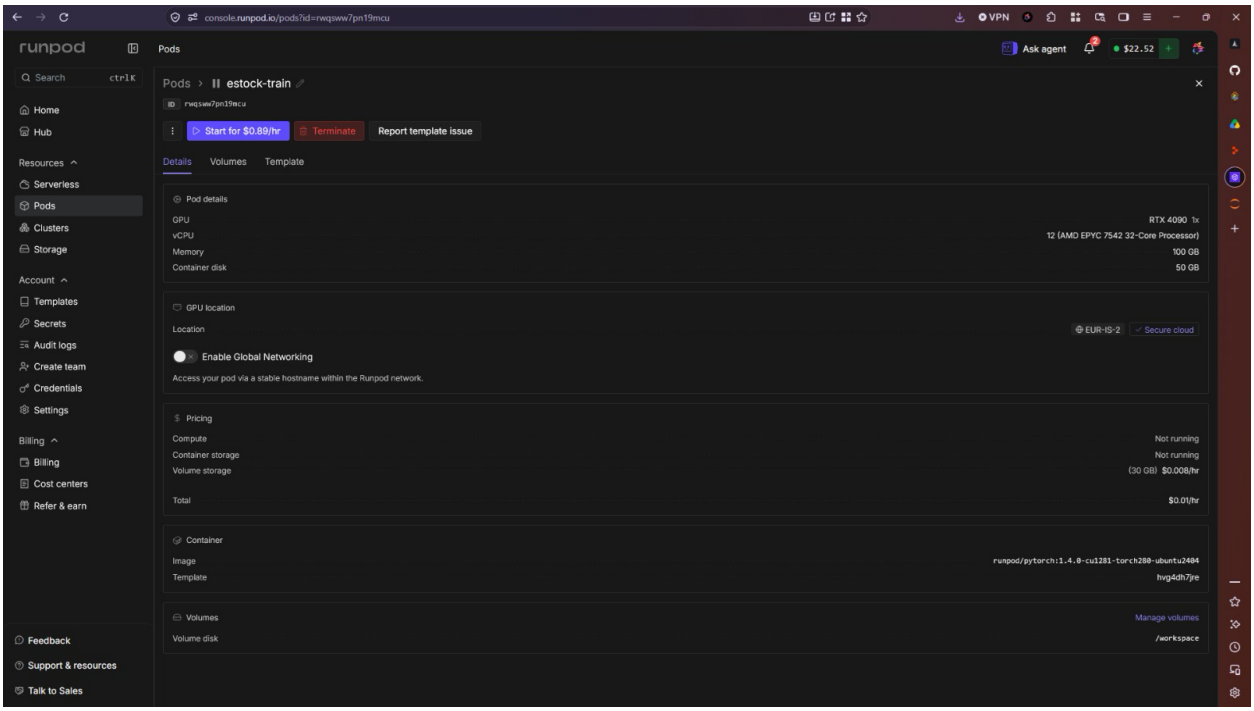
Recorded run identifier:

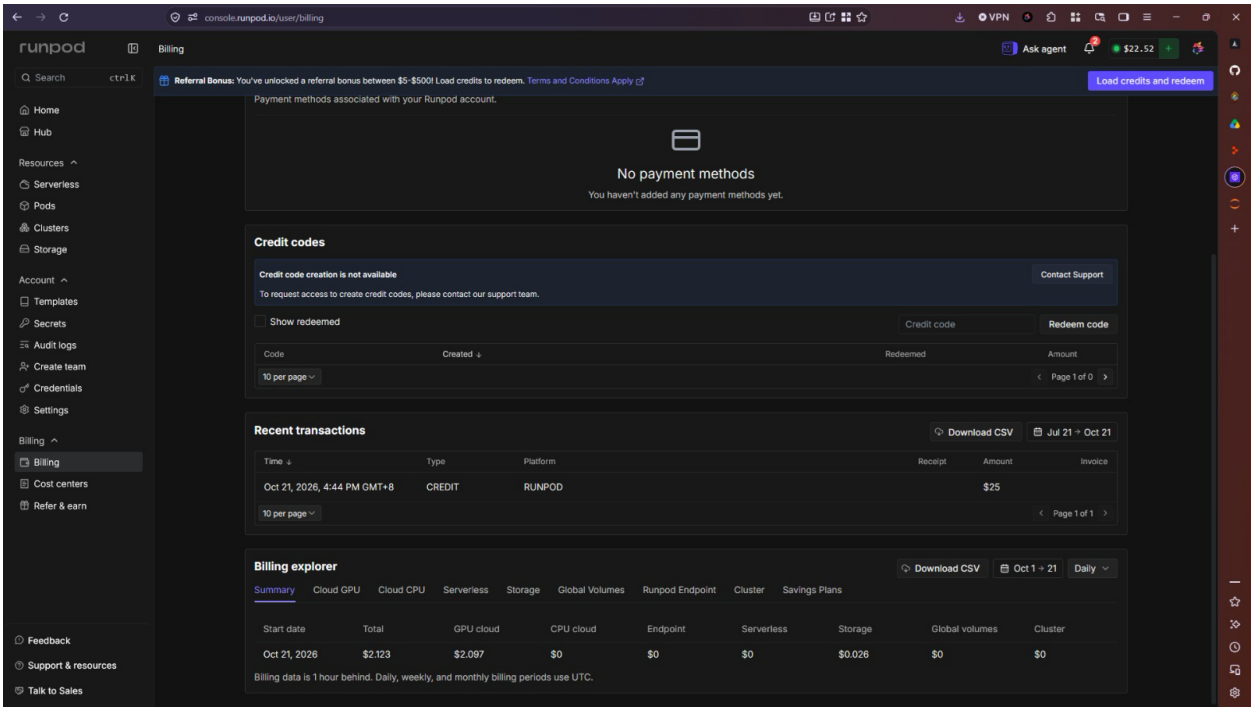
20261021T110320Z

Recorded experiment window:

21 October 2026, approximately 19:03:20–19:10:08 MYT.

This experiment window is not a declaration of total billable Pod uptime.





```json
{
  "run_id": "20261021T110320Z",
  "base_model": "autogluon/chronos-2",
  "base_revision": "60088152a34e242427b44c3100014473a0157d53",
  "fine_tune": "AutoGluon Chronos2 defaults (LoRA)",
  "prediction_length": 44,
  "validation_offsets_from_end": [
    110,
    96
  ],
  "test_offset_from_end": 44,
  "data_files": {
    "sales_history.csv": "1f844e0ef85670899f759e3dfdb1315d7d5324547f02c246cba8fd2e554d80ea",
    "estock_sales_daily_5sku_2020-01-01_to_2026-01-01.csv": "cde5b8da146517ada8ccdb64d05f23019313a4b87b3b90527f373a6f244f16ed"
  },
  "gpu": "NVIDIA GeForce RTX 4090",
  "versions": {
    "autogluon.timeseries": "1.6.3",
    "torch": "2.13.0+cu130"
  },
  "fit_seconds": {
    "validation": 141.1,
    "test": 132.2,
    "production": 130.5
  },
  "started_at": "2026-10-21T11:03:20+00:00",
  "finished_at": "2026-10-21T11:10:08+00:00"
}
```


Historical RunPod use remains declared even though the current receipt workflow uses Gemini and does not depend on the forecasting model.

Final PDF filename:

SDC2026_TooGoodToBeTrue_TechDeclaration.pdf